

Question 1.1. At least _____% of the people are between 25 and 65 years old. **(6 Points)**

93.75

Question 1.2. At most _____% of the people have ages that are not in the range 25 years to 65 years.
(6 Points)

6.25

Question 1.3. At most _____% of the people are more than 65 years old. **(6 Points)**

Hint: If you're stuck, try thinking about what the distribution may look like in this case.

3.125; The age distribution is unknown, and so are the proportion of ages either below 25 or above 65. So, I have assumed these distributions to be the same, meaning that the maximum proportion for both ages is equal to half of 6.25%, or 3.125%.

Question 2.2. Suppose the data science class decides to construct a 90% confidence interval instead of a 95% confidence interval, but they still require that the width of the interval is no more than 6% from left end to right end. Will they need the same sample size as in 2.1? Pick the right answer and explain further without calculation. **(6 Points)**

1. Yes, they must use the same sample size.
2. No, a smaller sample size will work.
3. No, they will need a bigger sample.

2; The 90% confidence interval requires a lower Z-score than a 95% confidence interval, meaning that a smaller sample size can still achieve the same width of the confidence interval.

Question 2.4. This shows that the percentage in a normal distribution that is at most 1.65 SDs above average is about **95%**. Explain why 1.65 is the right number of SDs to use when constructing a **90%** confidence interval. **(6 Points)**

A 90% confidence interval takes the middle 90% of the distribution of the data, meaning that the endpoints of the interval are the 5th and 95th percentiles of the distribution. Since 1.65 SDs above average is about 95% of the data, it means that the 5th and 95th percentiles are -1.65 and 1.65, respectively, making 1.65 the correct number of SDs for a 90% confidence interval.

Question 3.2. Why does the Central Limit Theorem (CLT) apply in this situation, and how does it explain the distribution we see above? We recommend reviewing [14.4](#) for a refresher on CLT. **(6 points)**

According to the Central Limit Theorem, the distributions of the sample means of larger sample sizes will eventually converge to a normal distribution. In this case, despite there being only two possible outcomes (yes and no), the proportions of votes of “yes” and “no” are similar in each of the many samples taken. So, the distribution of these samples are nearly normal. The Central Limit Theorem also allows us to use distribution’s properties to estimate potential confidence intervals for the proportion of “yes” votes.

